INCH-POUND
A-A-59867
17 May 2010

COMMERCIAL ITEM DESCRIPTION

HARNESS, SAFETY, FULL BODY, CROSSOVER

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description provides the requirements for the full body crossover style safety harness for use in hazardous fall operations onboard U.S. Navy surface ships, submarines, and shore activities.

2. CLASSIFICATION. The full body crossover style safety harness shall conform to the following sizes: X-Small, Small, Standard, Universal, and X-Large sizes (see figure 1 and table I).

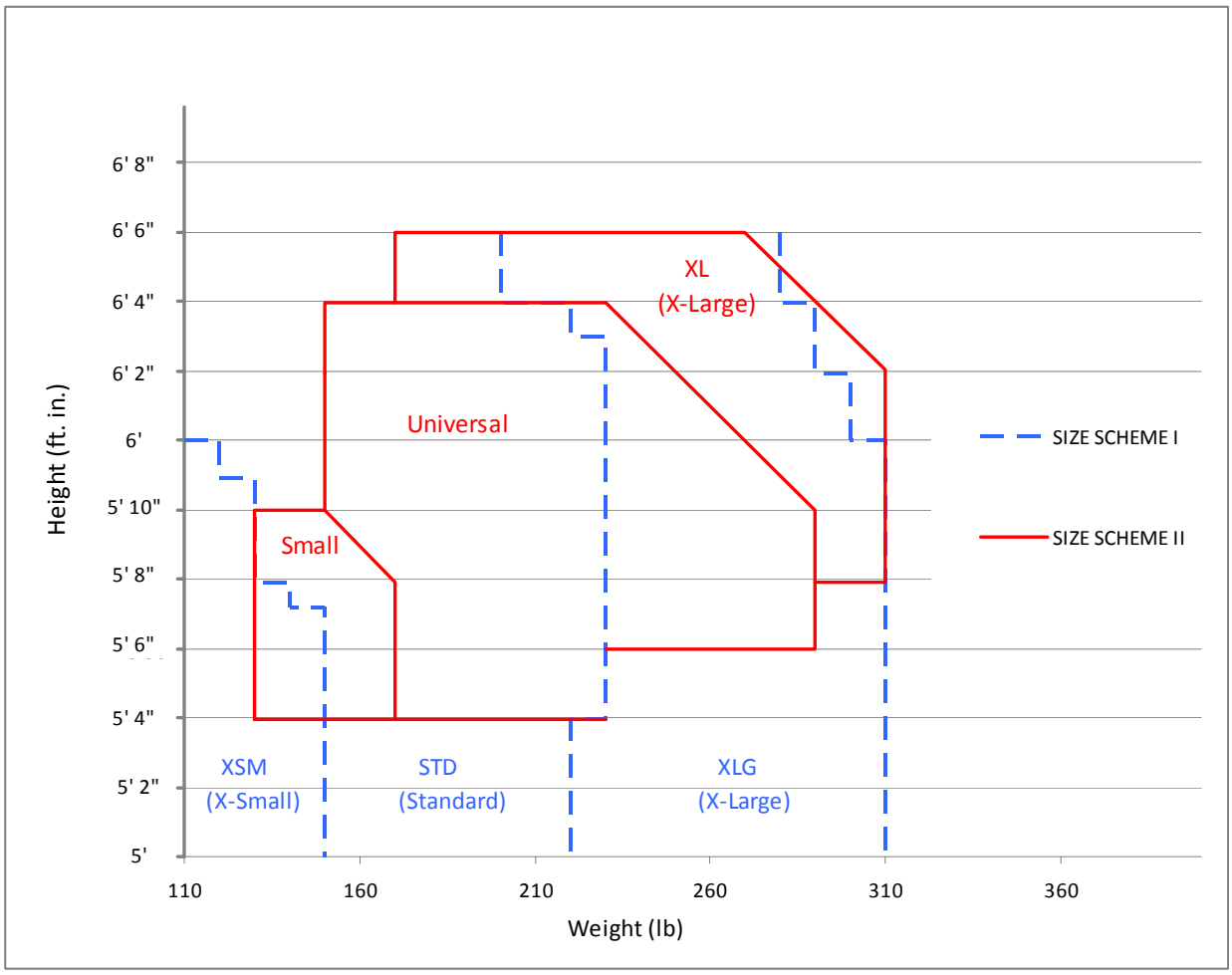
TABLE I. Harness sizes.

	Weight (lbs)					
Height	**<130**	**130 – 150**	**150 – 170**	**170 – 230**	**230 – 290**	**290 – 310**
Shorter than 5' 4"	XSM	XSM	STD	STD	XL	XL
Short (5' 4" – 5' 7")	XSM	XSM or SM	SM or STD	STD or Universal	Universal or XL	XL
Regular (5' 8" – 5' 11")	XSM	SM or STD	STD or Universal	STD or Universal	Universal or XL	XL
Tall (6' 0" – 6' 3")	STD	STD	STD or Universal	STD, Universal, or XL	Universal or XL	XL
Extra tall (6' 3" – 6' 6")	STD	STD	STD	STD or XL	XL	XL

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FSC 4240

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NOTES:

1. Size Scheme I: NSNs: 4240-01-545-7026 (X-Small), 4240-01-544-3376 (Standard), and 4240-01-544-3382 (XL).
2. Size Scheme II: NSNs: 4240-01-543-5503 (Small), 4240-01-543-5507 (Universal), and 4240-01-543-5520 (XL).

FIGURE 1. Harness sizes in graphic form.

3. SALIENT CHARACTERISTICS.

3.1 Design and construction. The full body crossover style safety harness shall be comprised of load bearing straps and hardware. The harness shall distribute the load from a fall over the wearer's skeletal structure. It shall be in accordance with ANSI Z359.1 and 29 CFR 1910, 1915, and 1926. The full body crossover style safety harness shall have the following features:

a. The harness shall be a one piece design, which has straps across the lower chest, over the shoulders, around the thighs, and across the pelvis as shown on figure 2.

b. The harness shall be designed to allow a person to properly don and adjust an appropriate sized harness based on the information in table I and figure 1 in less than three minutes. It shall also be able to be removed readily without becoming tangled.

c. The harness shall be compatible with the U.S. Navy Life Preservers, described in A-A-59767 and A-A-59766.

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d. The harness shall be designed for personnel weighing less than 310 pounds.

e. The harness shall have six circular forged D-rings. They shall be located as follows: one dorsal D-ring between the wearer's shoulder blades, one on each side at waist level, one at the chest level, and one on the top of each shoulder.

f. The harness shall have either one or two frontal pass through buckles and color coded straps to allow ease of use for donning. A harness with one frontal pass through buckle shall be yellow with black or navy blue leg straps. A harness with two pass through buckles shall be red with blue leg straps.

g. The harness shall be compatible with shock absorbing or restraint lanyards that are made in accordance with ANSI Z359.1.

3.1.1 Hardware. The hardware or connectors shall be finished free of scale, rust deposits, or foreign matter. The surface of the hardware shall have no burrs, pits, sharp edges, or rough surfaces. The material for the hardware shall be high tensile alloy steel produced by forging or stamping. The restrictions associated with any hardware shall be marked on the hardware or components, subsystems, and systems in which they are an integral part. A hardness of Rockwell C35 to 42 is required for all high tensile alloy connectors.

3.1.2 Straps. The straps shall be made from light, heat, ageing, and abrasion resistant continuous virgin polyamide filament yarns or equivalent. The minimum width of the straps shall be 1⅝ inches (41 millimeters) and the strap ends shall be finished to prevent fraying. The straps' breaking strength shall be more than 6000 pound force (lbf) (26.69 kilo Newton (kN)).

3.1.3 Thread and stitching. All threads shall be made of the same material as the polyamide or equivalent threads used to make the straps. The color of the threads shall contrast with the strap color for ease of inspection. Strap joints shall be sewn by lock stitching.

3.1.4 Labeling. Labeling shall be in accordance with ANSI Z359.1.

3.1.4.1 Legibility. The legibility and attachment of required markings shall last for the life of the harness. When pressure sensitive labels are used, they shall be in accordance with the provisions of UL 969.

3.1.4.2 General markings. The following shall be marked in English on the equipment in accordance with ANSI Z359.1: National Stock Number incorporated onto an existing label, part number and model designation, year of manufacture, manufacturer's name and logo, capacity rating, standard number "Z359.1", and warning to follow the manufacturer's instructions included with the equipment at time of shipment from the manufacturer.

3.1.4.3 Specific markings. The harness shall be marked to identify the following: the fiber used in the material construction, the size of the harness, the fall arrest attachment element(s), the purpose of any other attachment elements, and a caution to use the front mounted attachment element only for a maximum of two feet free fall distance.

3.1.4.4 General and specific instructions. General instructions shall be provided in accordance with ANSI Z359.1. Specific instructions shall explain the maximum free fall distance permitted; all the markings with appropriate elaborations and explanations; and donning, fitting, adjustment, and use information including warnings against improper methods of assembly.

3.2 Performance requirements.

3.2.1 Dynamic strength test. The dynamic strength test shall be conducted in accordance with ANSI Z359.1.

3.2.1.1 Drop test structure. The drop structure shall allow enough vertical and horizontal clearance for the test with no obstruction. The natural frequency of the drop test structure shall be 200 Hertz. The maximum elastic deformation of the anchorage under 2250 lbf (10 kN) should be 0.04 inch.

3.2.1.2 Test torso. Test drop weight shall have a form of a human torso with dimensions in accordance with ANSI Z359.1. The torso shall be rigid, have hardwood surface, and weigh 200±2 pounds (100±1 kg).

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3.2.1.3 Test lanyard. The test lanyard shall be made from Type 302 stainless steel, 7 by 19 aircraft cable construction in accordance with MIL-DTL-83420. The cable diameter shall be ⅜ inch (9.5 millimeters) and lanyard eyes shall be formed by flemish eye splices and secured with swaged fittings.

3.2.1.4 Quick release mechanism. The test torso and harness ensemble shall be released from the drop height via remotely operated quick release mechanism which synchronously triggers the data recording device. The data for the entire history of fall arrest shall be captured.

3.2.1.5 Testing and evaluation. The test torso and harness assembly shall free fall from a height of 3.3 feet (1 meter) and 12 inches (305 millimeters) maximum lateral distance from the anchorage via quick release mechanism. After the drop, the assembly shall be suspended for 5 minutes while connected to the fall arrest system from the dorsal ring of the harness. The harness shall endure the dynamic forces of the drop and suspension period without breaking. The angle between the torso vertical centerline and vertical shall be no more than 30 degrees after the torso comes to rest. This test shall be conducted twice, once feet-first and once head-first. A new harness may be used for each test.

3.2.2 Dynamic strength test (frontal attachment). The dynamic strength test (frontal attachment) shall be conducted and evaluated the same way as the dynamic strength test, except that the test lanyard of the fall arrest system shall be connected to the frontal D-Ring of the harness.

3.2.3 Strength of d-rings, o-rings, and oval rings. The strength of this hardware shall be tested in accordance with ANSI Z359.1. A static tensile test load of 5000 lbf (22.2 kN) shall be internally applied to the hardware for a minute. The hardware shall stay intact to pass the test.

3.2.4 Strength of buckles and adjusters. Oval rings used as adjusters or other adjusters shall be subjected to Tensile Testing of Buckles and Adjusters test in accordance with ANSI Z359.1. The specimen shall be exposed to a tensile static force of 4000 lbf (17.7 kN) for a minute. It shall take one minute or more for the test machine to bring up the test load to avoid dynamic effects. The specimen shall not break to pass the test.

3.2.5 Proof loading of d-rings and oval rings. A tensile load of 3600 lbf (16 kN) shall be brought up to the specimen in more than one second and maintained for at least two seconds in accordance with ANSI Z359.1. Inspection after testing shall not reveal any cracking, breaking, or permanent deformation visible to the unaided eye.

3.2.6 Tensile strength of straps. Straps shall not break under 6000 lbf (26.69 kN) test load when tested in accordance with ASTM D6775 Standard Test Method for Breaking Strength and Elongation of Textile Webbing, Tape and Braided Material.

3.2.7 Static strength of the full body harness. The standard 220±2 pounds (100±1 kilograms) torso shaped and hardwood surfaced rigid test weight shall be properly fitted in the full body harness in accordance with ANSI Z359.1. The harness shall then be attached to the lower and upper jaws of the testing machine by the torso crotch ring and harness dorsal D-ring, respectively. The location of the buckles and adjusters shall be marked before testing in order to measure slippage later. The load shall be increased up to 6000 lbf (26.69 kN) at a rate of 2 inches/min (51mm/min) and maintained for a minute. Harness shall hold the test torso during the test and no element should break. Adjustable connectors shall not slip more than 1 inch (25 millimeters). Buckle and eyelet type of construction shall not tear a distance greater than that to the adjacent eyelet.

3.2.8 Static strength testing of full body harness (front mounted attachment). The same test for static strength for the full body harness shall be used, except the load to the harness frontal D-ring shall be increased to only 3600 lbf (16 kN) in accordance with ANSI Z359.1. The same procedures to make a pass or fail decision shall be followed.

3.2.9 Corrosion resistance. All the hardware shall pass a 48 hours salt spray test in accordance with ASTM B117. There shall be no sign of corrosion or red rust formation when inspected by an unaided eye.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. Manufacturer shall show that they have made similar type harnesses which meet ANSI Z359.1 and 29 CFR 1910, 1915, and 1926.

5.3 First article. One full body crossover harness shall be subjected to first article inspection in accordance with 3.1 and 3.2. Results shall be provided as specified in the contract or order (see 7.2.b).

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 ANSI. American National Standards Institute (ANSI) standards are available from International Safety Equipment Association, 1901 N. Moore Street, Suite 808, Arlington, VA 22209-1762 or online at http://webstore.ansi.org/.

7.1.2 ASTM. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.1.3 CFR. The Code of Federal Regulations (CFR) may be obtained from Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20401 or online at www.gpoaccess.gov/index.html.

7.1.4 Defense standards. Defense standards may be obtained from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil.

7.1.5 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

7.1.6 UL. Underwriters Laboratories, Inc. standards are available from COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515 or online at http://www.ul.com.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number.
b. National stock numbers.
c. Product conformance provisions, including first article test results (see 5.2).
d. Quantity required.
e. Packaging requirements.

7.3 National stock numbers (NSNs). The following is a list of NSNs assigned corresponding to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	ITEM
4240-01-545-7026	Size X-Small Safety Harness
4240-01-543-5503	Size Small Safety Harness
4240-01-544-3376	Size Standard Safety Harness
4240-01-543-5507	Size Universal Safety Harness
4240-01-544-3382 or 4240-01-543-5520	Size X-Large Safety Harness

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7.5 Key words.

Fall Protection

Hardware

Straps

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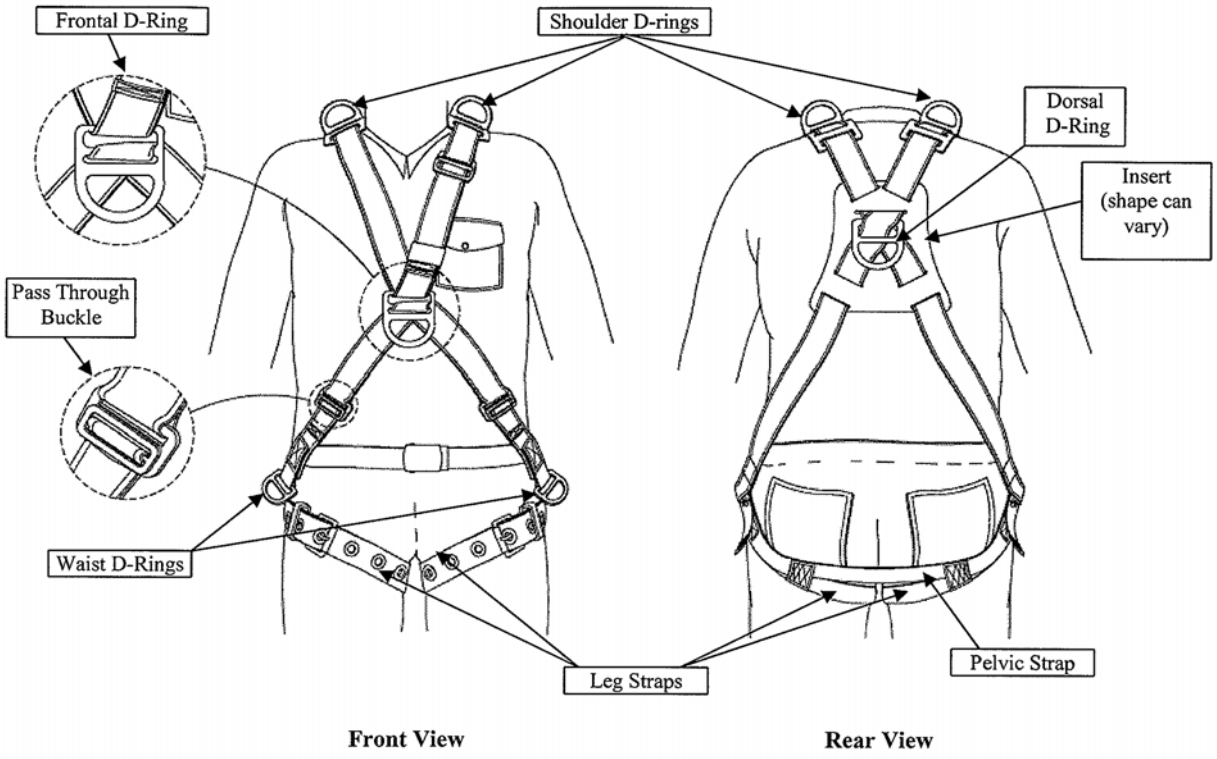


FIGURE 2. Full body, crossover, safety harness (two front pass through buckles version shown).

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MILITARY INTERESTS

Custodians:
Army – EA
Navy – SH
Air Force – 99

Review Activities:
Army – TE
Navy – AS
Air Force – 03, 84

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 4240-2009-001)

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